

Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Outline

Intro

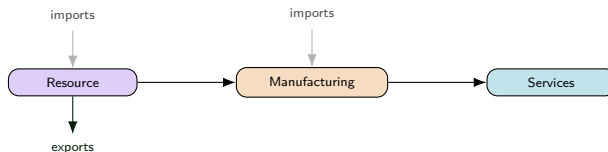
Model

Results

Conclusion

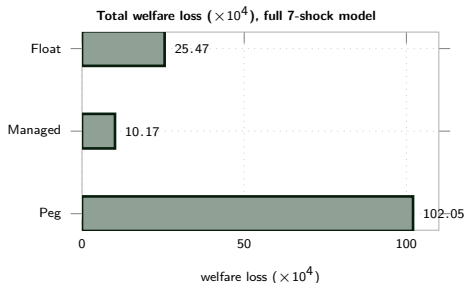
Motivation

- Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index
- Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**
- **This paper: both together** — multi-sector network + FX



Questions: Optimal FX regime? Does the DC index survive a 2nd cost-push channel?

Headline Result



- **Managed float dominates** both corners
- **Peg dominated:** zero autonomy, UIP shock passes straight through
- **Driver:** persistence (near-unit-root NFA), not volatility

Units: consumption-equivalent loss, $\times 10^{-4}$. Calibrated to Chile's real national input-output table (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. year 2023) — full calibration and sources on the next two slides. Solved as the genuine nonlinear model (Dynare), not linearized.

Literature

Network side (closed economy):

- **Rubbo (2024)** — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
- **La'O & Tahbaz-Salehi (2022)** — optimal MP in production networks
- **Pasten, Schoenle & Weber (2020)** — sectoral rigidity heterogeneity drives MP transmission

Open-economy side (no network):

- **Galí & Monacelli (2005)** — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT-adjustment channel)
- **Fanelli & Straub (2021)** — FX intervention as a *second* instrument, separate from i_t
- **Schmitt-Grohé & Uribe (2003)** — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
- **Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019)** — pass-through, dominant currency, IO spillovers

Gap: network + open economy + FX together — no one has this.

- Same ranking as Galí–Monacelli, **different channel** (risk-premium, not ToT)
- Managed's dominance echoes Fanelli–Straub's two-instrument logic
- Sector heterogeneity = FX analogue of Pasten–Schoenle–Weber

This Paper

- Extend Rubbo → SOE: imported inputs Ω^F , UIP, debt-elastic premium
- Ask: optimal FX regime? DC index survive with 2 cost-push channels?
- **Not just Galí-Monacelli with sectors**: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new
- Calibrate: 3-sector oil exporter; solve **nonlinear** model directly (not hand-linearized)

Model: Households

$$\begin{aligned} \max \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \quad \text{s.t.} \quad & P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^* \\ & C_t^{-\gamma} = \beta(1+i_t) \mathbb{E}_t \left[C_{t+1}^{-\gamma} / \Pi_{t+1}^C \right], \quad W_t / P_t^C = C_t^\gamma L_t^\varphi \\ \text{UIP (derived):} \quad & (1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t] \end{aligned}$$

C, L : consumption, labor
 ψ : debt-elastic premium

β, γ, φ : discount, RRA, inv. Frisch
 B^* : foreign bond

W, i : wage, dom. rate
 P^C : CPI

S, i^* : ER, foreign rate
 Π_t : profits (rebated)

Model: Firms — Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \quad \Rightarrow \quad MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A : output, TFP L, X, M : labor, dom./imp. inputs Ω_{ij}^H : dom. IO share Ω_i^F : import cost share
 α_i : labor share MC_{it} : nominal marg. cost P_t^{F*} : foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Model: Firms — Calvo Pricing

Reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^\varepsilon X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^\# / P_{it} = \frac{\varepsilon}{\varepsilon-1} X1_{it} / X2_{it}, \quad 1 = (1-\delta_i)\Pi_{it}^{\varepsilon-1} + \delta_i \left(P_{it}^\# / P_{it}\right)^{1-\varepsilon}$$

δ_i : Calvo reset prob. (primitive) $\hat{\delta}_i$: derived slope $P_{it}^\#$: reset price ε : elasticity (markup $\varepsilon/(\varepsilon-1)$)

$$\hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)} \text{ — derived slope, enters DC weights.}$$

Model: Open Economy

$$\underbrace{P_t^{FH} = S_t P_t^{F*}}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \quad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}} \right)^{-\theta^*}$$

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}} \left(C_t^H + EX_t \right) + \sum_j \Omega_{ji}^H \frac{MC_{jt}}{P_{it}} \frac{Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled)

β_i^H : cons. share, sector i

EX, IM : exports, imports

Float: i_t Taylor rule, S_t free

θ^* : export elasticity

D^* : foreign demand shifter

Peg: $S_t = \bar{S}$, i_t residual

Model: Policy Regimes and Shocks

Regime	Rule	Exchange rate
Float	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual
Managed	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Shocks (AR(1)): a_{it} (TFP, $\rho=0.90$) p_t^{F*} (import price, $\rho=0.85$) D_t^* (foreign demand, $\rho=0.80$)

Theory: Regime Comparison — Predictions

	Float	Peg	Managed
Exchange rate	endogenous	fixed	partial
FX cost-push absorbed by	S_t	π_i, y	split
Monetary independence	full	none	full

Naive: float best. **Overtuned** — see Welfare section.

Does the DC Index Survive a Second Cost-Push Channel?

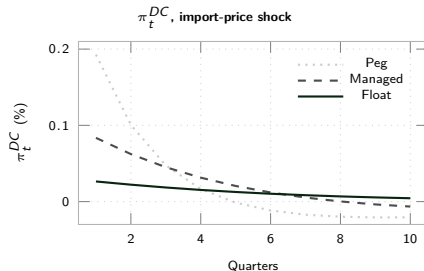
Closed economy (Rubbo): 1 channel (TFP) $\Rightarrow$ unique DC index

$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}$$

Open economy: 2 channels (TFP + FX) $\Rightarrow$ need *both*, same weights ϕ

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

- **Answer:** no, generically
- This calibration: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ — close, but not constant
- **Fix:** flexible import pricing ($\hat{\delta}_M=1$) restores π_t^{DC}



Full DC-index algebra, proof, and knife-edge test:
Appendix.

Calibration: Chile, Real IO Data

Parameter	Value	Parameter	Value
β	0.99	$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604
γ, φ	1.00, 2.00	$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070
η	1.50	$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744
ε	8.00	$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16
θ^*	2.00	ψ	0.020
$\beta^{F,tot}$	0.10	ϕ_π, ϕ_y, ϕ_s	1.50, 0.50, 0.30

Dense domestic IO matrix (Resource, Manufacturing, Services order), all 9 entries used — no triangular-network assumption:

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}, \quad \text{Export share} = (0.602, 0.180, 0.036)$$

- Resource = mining/copper (Chile's export engine, 60% of own output exported) → Manuf. (mid-chain) → Services (largest value added, sticky)
- $\alpha_i, \Omega_{ij}^H, \Omega_i^F, \beta_i^H$: **data** (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. year 2023) — sector concordance & full build script in `data_calibration/build_chile_calibration.py`
- $\delta_1, \delta_2, \delta_3$ (Calvo reset prob.): **literature**, not data-identifiable — euro-area monthly price-change frequencies (Dhyne et al. 2005/ECB IPN: energy 78.0%, unprocessed food 28.3%, processed food 13.7%, non-energy industrial goods 9.2%, services 5.6%), converted to quarterly via $\delta_q = 1 - (1 - f_{mo})^3$;

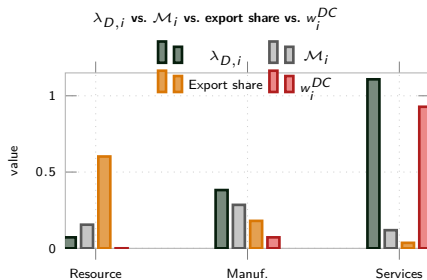
Calibration: Network Properties

$$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i$$

$$\mathcal{M}_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i$$

$$w_i^{DC} \propto \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

$\lambda_{D,i}$: Domar weight $\mathcal{M}_i$: import centrality w_i^{DC} : DC weight

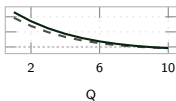


- FX exposure runs **opposite** to Domar weight
- Resource: small weight (0.07), most FX-exposed ($\mathcal{M}_i = 0.16$) — and *is* the export engine (60% of own output exported, copper)
- Services: dominates output ($\lambda_D = 1.11$), FX-exposed only **indirectly**
- w_i^{DC} is essentially a Services index (0.93)

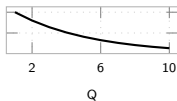
IRFs: Foreign Risk-Premium (UIP) Shock — the Key Mechanism

The shock that decides the ranking: 75% of Peg's welfare loss (Chile calibration).

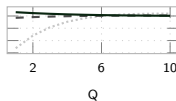
Exch. rate S_t (% , dep.>0)



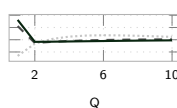
Shock: risk premium RP_t (%)



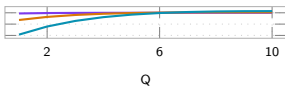
Output gap $\tilde{y}_t$ (%)



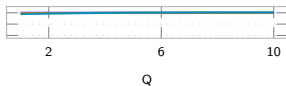
CPI infl. π_t^C (%)



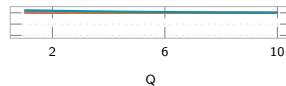
Sectoral output gap, Peg (%)



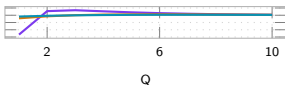
Sectoral output gap, Managed (%)



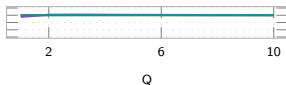
Sectoral output gap, Float (%)



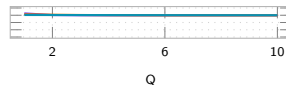
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

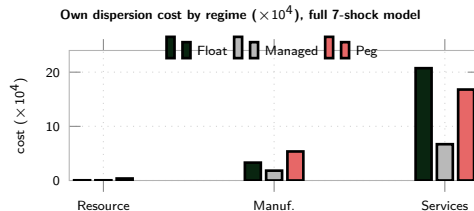


Sectoral inflation, Float (%)



Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

Network Exposure & Sector Welfare Preferences

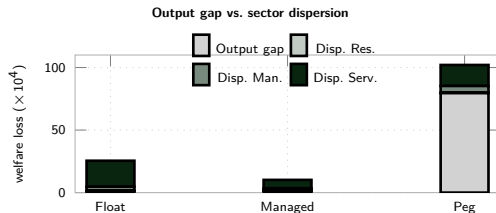
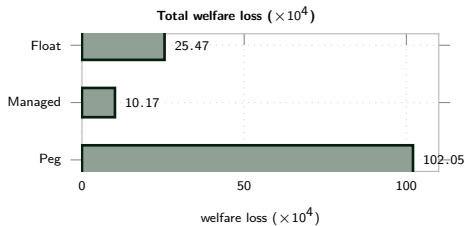


Exposure $\rightarrow$ pass-through $\Gamma_i = \frac{[\Delta(1-\Omega\Delta)^{-1}\omega_F]_i}{\kappa_i}$ and welfare weight $\kappa_i = \lambda_{D,i}\varepsilon \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ (Chile, real Ω^H, Ω^F , literature $\hat{\delta}_i$):

	$\mathcal{M}_i$	Export share	Γ_i	κ_i
Resource	0.155	0.602	1.091	0.07
Manuf.	0.285	0.180	0.001	21.5
Services	0.119	0.036	<0.0001	276.3

- Under the literature-sourced Calvo stickiness ($\hat{\delta}_3 = 0.03$, Services extremely sticky), κ_i is overwhelmingly concentrated in Services — so FX pass-through, though present in all sectors via $\mathcal{M}_i$, barely matters for Services' welfare cost *relative to its own weight*: almost all of Γ_i funnels through Resource
- Managed dominates every sector.** Peg is costliest for Resource (0.37) and Manuf. (5.36); but for Services, **Float is costliest** (20.73 > Peg's 16.79) — the near-unit-root S. under Float feeds the stickiest sector's marginal cost for a very

Welfare: Headline & Persistence

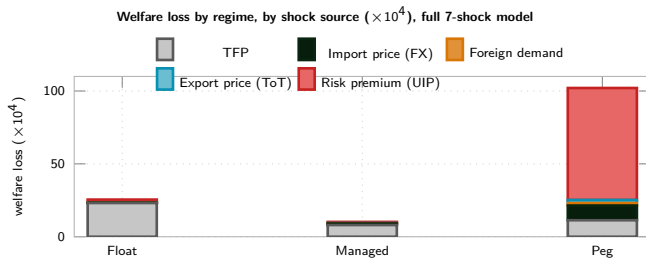


$$\mathcal{W} = \frac{1}{2} \left[(\gamma + \varphi) \text{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \text{Var}(\pi_{it}) \right] \text{ — Peg dominated on output gap \& dispersion, every sector.}$$

Why: persistence, not volatility.

- Float's S_t (near-unit-root NFA) feeds every sector's MC, even for domestic shocks
- Peg's output-gap term explodes (79.5) because it has zero monetary autonomy to absorb the risk-premium shock
- Slow decay, not larger impact, drives Float's Services dispersion above Managed's

Shock Decomposition of Welfare Cost



- **Risk-premium/UIP dominates Peg** (75% of its total)
- **Ranking:** Managed < Float $\ll$ Peg — Peg no longer competitive

Caveat: 1st-order welfare — this channel is where 2nd-order could move most (see Conclusion).

Mechanisms: Why Each Regime Costs What It Costs

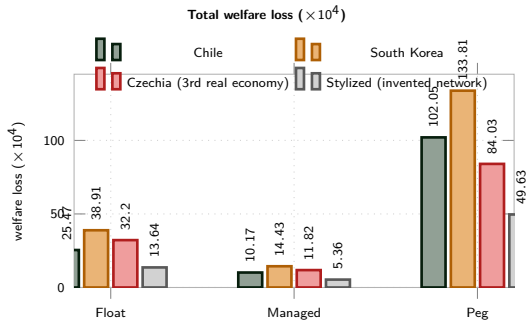
Regime	Cost channel	What sets the magnitude
Float	S_t 's near-unit-root persistence (NFA state) feeds $\Gamma_i \Delta e_t$ into every sector's MC, even for shocks with no FX content — pure dispersion cost	Persistence of debt-elastic premium (ψ); Γ_i (import exposure); κ_i of the most central/sticky sector (Services: 20.7 of 25.5)
Peg	Zero monetary autonomy: output-gap term explodes (79.5 of 102.1, $\approx 78\%$) on the UIP shock; $\Delta e_t \equiv 0$ <i>relocates</i> , not removes, adjustment into upstream P^H	Absence of ϕ_π, ϕ_y ; real vs. nominal share of the shock; upstream sector's Domar weight
Managed	$\phi_S \Delta \log S_t$ damps $\Gamma_i \Delta e_t$ without fully re-imposing peg's relocation cost — dominates every sector at once	Interior optimal $\phi_S^* \approx 0.20$: too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ autonomy lost

Two forces, always in tension: Γ -channel favors peg (switches off $\Gamma_i \Delta e_t$); **relocation** channel favors float (peg's fixed e_t forces the same adjustment into upstream domestic prices instead). Which wins for a sector depends on Γ_i vs. κ_i (Network Exposure slide); managed is the only regime improving on *both* corners for every sector.

Analytical or simulated? Both.

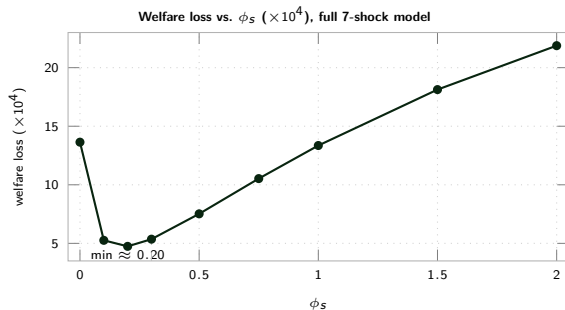
- **Channels — closed-form:** $\Gamma_i = [\tilde{A}\omega_F]_i / \kappa$ (pass-through), $\kappa_i = \lambda_{D,i} \varepsilon^{\frac{1-\delta_i}{\delta_i}}$ (welfare weight), DC-survival $\phi^\top \mathcal{V} = \mathbf{0}^\top$ and $\phi^\top \Gamma = \mathbf{0}$ — from primitives alone, no model solve.
- **Magnitudes/ranking — need simulation:** $\text{Var}(\pi_{it}), \text{Var}(\tilde{y}_t)$ come from the full linear RE solution (order-1 Dynare) — no closed form once $\phi_S > 0$ or Ω^H dense. Managed < Float $\ll$ Peg and $\phi_S^* \approx 0.20$ are simulated, but robust across all four calibrations.

Robustness: South Korea, Czechia & the Stylized Baseline



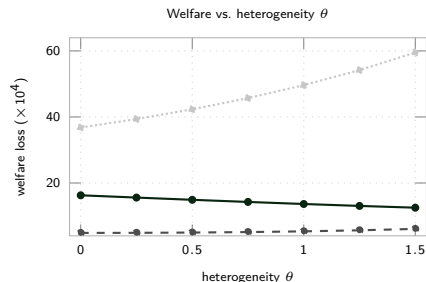
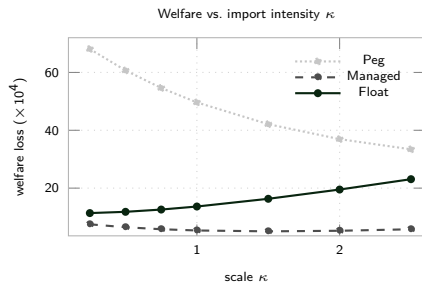
- **South Korea:** diversified manufacturing exporter (Bank of Korea, 2022 IO Tables), no natural-resource story to lean on — deeper manufacturing import intensity than Chile ($\Omega_{\text{manuf}}^F = 0.24$ vs. 0.19)
- **Czechia:** a third real IO calibration (Eurostat naio_10_cp1700, 2022) — deepest manufacturing import intensity of the three ($\Omega_{\text{manuf}}^F = 0.32$, EU/German supply chains), same concordance and pipeline as Chile/Korea
- **Stylized:** the original hand-picked triangular Ω^H (Rubbo-style, no data), kept only as a check that the qualitative result isn't an artifact of a specific network shape
- **Ranking survives in all four:** Managed < Float < Peg

Generalization: Optimal FX Stabilization ϕ_s



- **Why interior:** too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ loses autonomy
- $\phi_s=0 \rightarrow 0.20$: loss 13.64 $\rightarrow$ 4.74
- **Peaked, not flat:** 0.20 beats calibrated 0.30 by $\approx 13\%$
- Beyond ≈ 0.2 : rises fast, back above Float by $\phi_s=1$

Generalization: Robustness

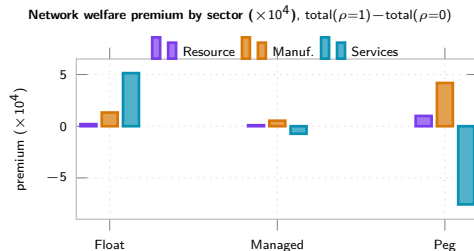
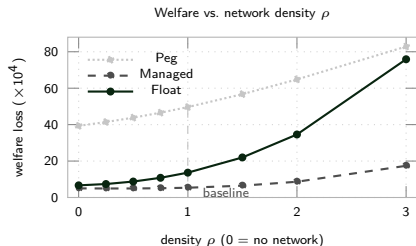


Managed < Float < Peg robust, but the gap moves in **opposite directions**:

- κ (import scale): Peg falls, Float rises — **converge**
- θ (concentration): Peg rises, Float falls — **diverge**

Isolating the Network Channel

New sweep: scale domestic I-O links OH_{21}, OH_{32} by ρ , holding ω_F & stickiness fixed. $\rho=0$: no domestic network. $\rho=1$: baseline.



- **Managed is almost network-neutral:** $\rho=0 \rightarrow 1$ adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg — $\phi_S > 0$ absorbs network amplification, not just direct FX cost-push
- **Denser network $\Rightarrow$ corners diverge from Managed:** by $\rho=3$, Float (75.9) catches Peg (82.9), both $\gg$ Managed (17.4) — **integrated economies should avoid either corner**
- **Burden lands on different sectors by regime:** Peg/Managed hit upstream Resource/Manuf., Services gains (−7.56/−0.73); Float hits Services most (+5.14) — the network transmits FX cost-push *indirectly*

Conclusion

- **Managed float optimal:** 5.36 vs. Float 13.64, Peg 49.63 ($\times 10^4$)
- **Driver:** risk-premium/UIP dominates Peg (81%); TFP persistence dominates Float & Managed
- **Optimal** $\phi_s \approx 0.20$: sharply peaked — over-stabilizing S_t is costly
- **Ranking robust** to κ, θ, ρ (network density); the *gap* is not
- **Network channel isolated:** $\Omega^H=0 \rightarrow$ baseline adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg — **denser networks favor Managed most**, and reallocate the network's sectoral burden (Services gains under Peg, loses under Float)
- **Vs. literature:** same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)
- **Caveat:** 1st-order welfare; UIP wedge & near-unit-root NFA are new distortions
- **Next:** order-2 solution

Appendix: DC Index — Definition and Insulation

Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$
rank-deficient $\Rightarrow$ unique DC index:

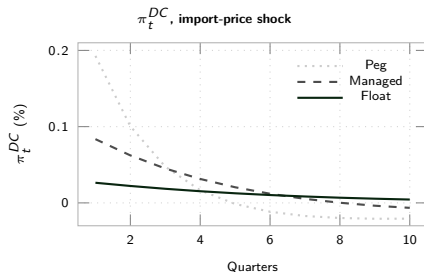
$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}, \quad w_i^{DC} \propto \lambda_{D,i} \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Open economy: 2 channels (TFP + FX):

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

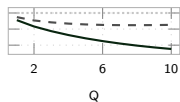
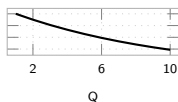
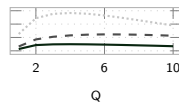
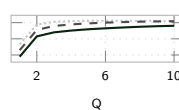
Knife-edge test, this calibration: $\Gamma \propto \mathcal{B}$ would save the DC index (Section 5.1). Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i/\mathcal{B}_i$ shrinks $\approx 2\times$ per sector (0.43, 0.18, 0.08), not constant. Breakdown is real, not a numerical artifact of an near-knife-edge calibration.

Fix: import nodes flexible ($\hat{\delta}_M=1 \Rightarrow w_M=0$) $\Rightarrow$ DC well-defined again by construction. This is what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown at right.

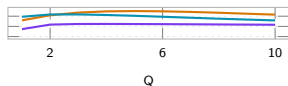


Float: smallest on impact (0.026%) & most persistent;
Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

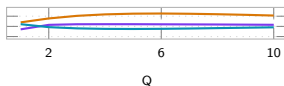
IRFs: Resource TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{1t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

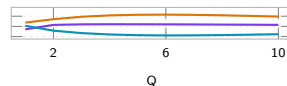
Sectoral output gap, Peg (%)



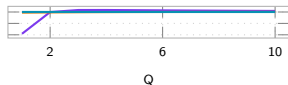
Sectoral output gap, Managed (%)



Sectoral output gap, Float (%)



Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

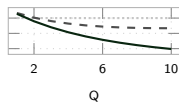
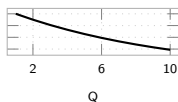
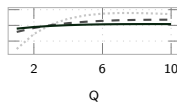
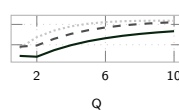


Sectoral inflation, Float (%)

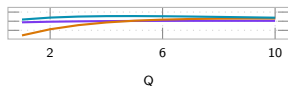


Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

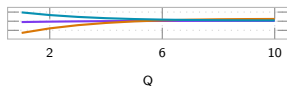
IRFs: Manufacturing TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{2t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

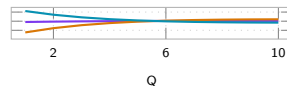
Sectoral output gap, Peg (%)



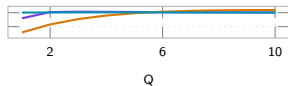
Sectoral output gap, Managed (%)



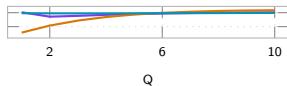
Sectoral output gap, Float (%)



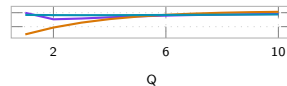
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

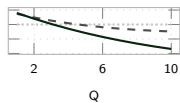
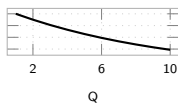
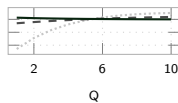
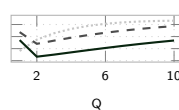


Sectoral inflation, Float (%)

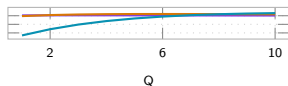


Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

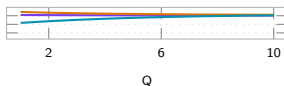
IRFs: Services TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{3t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

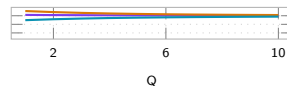
Sectoral output gap, Peg (%)



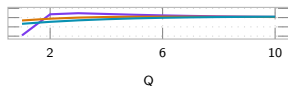
Sectoral output gap, Managed (%)



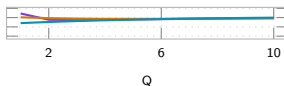
Sectoral output gap, Float (%)



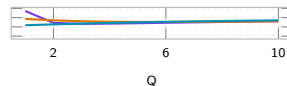
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

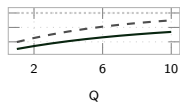
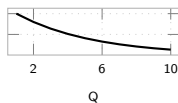
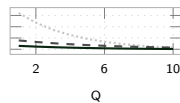
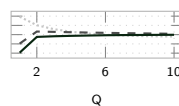


Sectoral inflation, Float (%)

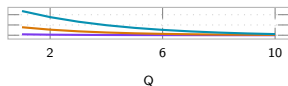


Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

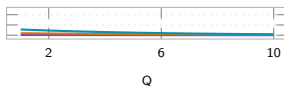
IRFs: Foreign-Demand Shock

Exch. rate S_t (% , dep.>0)Shock: foreign demand D_t^* (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, Peg (%)



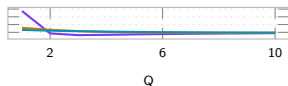
Sectoral output gap, Managed (%)



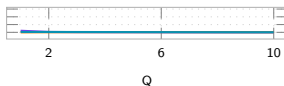
Sectoral output gap, Float (%)



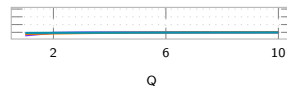
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

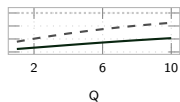
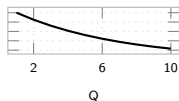
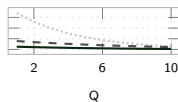
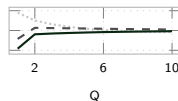


Sectoral inflation, Float (%)

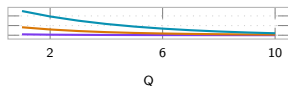


Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.

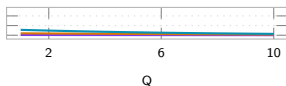
IRFs: Export-Price / ToT Shock

Exch. rate S_t (% , dep.>0)Shock: export price P_t^X (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

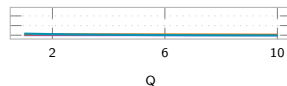
Sectoral output gap, Peg (%)



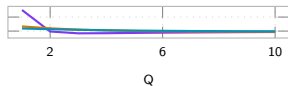
Sectoral output gap, Managed (%)



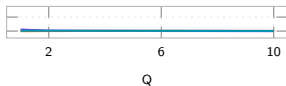
Sectoral output gap, Float (%)



Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

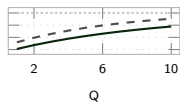
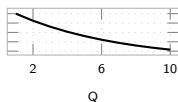
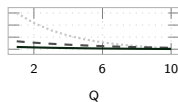
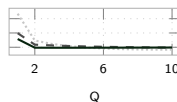


Sectoral inflation, Float (%)

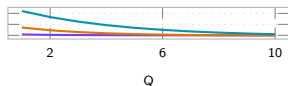


Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.

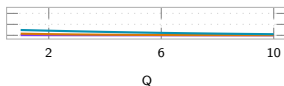
IRFs: Import-Price Shock (Backup)

Exch. rate S_t (% , dep.>0)Shock: import price P_t^{F*} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, Peg (%)



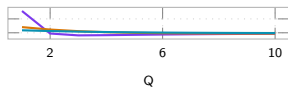
Sectoral output gap, Managed (%)



Sectoral output gap, Float (%)



Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)



Sectoral inflation, Float (%)



Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.